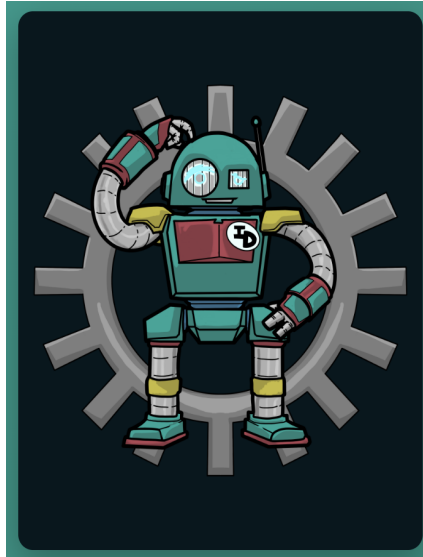




Code a Tank Drive!

FTC Team #22738 • Indecisive Devices

Today you are going to program a robot to drive using Java and Android Studio! ■

Java Survival Guide

Semicolon ;

Java uses semicolons like periods at the end of a sentence.

Curly Brackets { }

These group code together like LEGO bins.

Variables

Variables are labeled boxes that store information.

Methods

`init()` runs once when the robot starts.

`loop()` repeats over and over while driving.

Step 1: Type the Motor Setup Code

■ Robot Coach Says:

Inside the `public void init() { }` method, type these 4 lines EXACTLY as shown below.

This connects your code to the real motors on the robot.

```
frontLeftDrive = hardwareMap.get(DcMotor.class, "FL");  
rearLeftDrive = hardwareMap.get(DcMotor.class, "RL");  
frontRightDrive = hardwareMap.get(DcMotor.class, "FR");  
rearRightDrive = hardwareMap.get(DcMotor.class, "RR");
```

What is hardwareMap.get()?

It is like telling the robot:

'Go find the motor named FL and connect it to this code variable.'

Step 2: Reverse the Left Motors

Still inside the `init()` method, type these lines below the motor setup code.

```
frontLeftDrive.setDirection(DcMotorSimple.Direction.REVERSE);
rearLeftDrive.setDirection(DcMotorSimple.Direction.REVERSE);
```

Some motors are mounted backwards on the robot. REVERSE makes all wheels move forward together instead of fighting each other.

Step 3: Read the Controller Joysticks

Now go inside the `public void loop() { }` method.

Type these lines EXACTLY as shown.

```
double leftPower = -gamepad1.left_stick_y;
double rightPower = -gamepad1.right_stick_y;
```

The joysticks send numbers from -1 to 1.

These numbers become motor power values.

Step 4: Send Power to the Motors

Still inside the `loop()` method, type these lines below the joystick code.

```
frontLeftDrive.setPower(leftPower);
rearLeftDrive.setPower(leftPower);
frontRightDrive.setPower(rightPower);
rearRightDrive.setPower(rightPower);
```

Step 5: Add Telemetry Messages

Telemetry lets the robot talk back to the Driver Hub screen.

Type these lines below the motor power code.

```
telemetry.addData("Left Power", leftPower);
telemetry.addData("Right Power", rightPower);
```

Step 6: Build Your Code ■

Click the **hammer/build icon** in Android Studio.

This checks your code for mistakes before running the robot.

Common Errors: Missing semicolon ; Misspelled variable names Uppercase/lowercase mistakes (Java is CASE SENSITIVE) Missing brackets { } ■ **Robot Coach Says:**

Even professional programmers get syntax errors. The trick is staying calm and reading the error messages one at a time.

Complete Tank Drive Code

```
@Override
public void init() {
    frontLeftDrive = hardwareMap.get(DcMotor.class, "FL");
    rearLeftDrive = hardwareMap.get(DcMotor.class, "RL");
    frontRightDrive = hardwareMap.get(DcMotor.class, "FR");
    rearRightDrive = hardwareMap.get(DcMotor.class, "RR");

    frontLeftDrive.setDirection(DcMotorSimple.Direction.REVERSE);
    rearLeftDrive.setDirection(DcMotorSimple.Direction.REVERSE);
}

@Override
public void loop() {
    double leftPower = -gamepad1.left_stick_y;
    double rightPower = -gamepad1.right_stick_y;

    frontLeftDrive.setPower(leftPower);
    rearLeftDrive.setPower(leftPower);
    frontRightDrive.setPower(rightPower);
    rearRightDrive.setPower(rightPower);

    telemetry.addData("Left Power", leftPower);
    telemetry.addData("Right Power", rightPower);
}
```

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